Fig. 3: ADXL355 accelerometer sensor

PARTS LIST	
Semiconductors:	
Board1	- Arduino Uno
Accelerometer	- ADXL335
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 470-ohm
R2	- 10-kilo-ohm
Miscellaneous:	
S1	- Push-to-on switch
Cable	- Arduino USB cable

The game is designed on pygame library that shows the images of birds and the gun with the entire graphical interface. Communication between Python code and Arduino is done using serial communication; therefore you need to add your COM port number after the program starts. This will prompt you for a confirmation. Press y to start the game. The program screen will appear along with a circle.

Birds or ducks appear at random locations with random directions. Directions are depicted using numbers, where 1 represents north, 2 represents north-east and moving in clockwise direction to 8, which represents north-west.

At every 200ms, a 4-byte data is sent from Arduino that includes a start byte (*), followed by the location of x and y coordinates scaled between 0 and 20, and ending with 0 or 1, where 1 depicts that a gun was fired using S1.

Byte 1 is sent continuously so that data is not lost. After Python code detects the firing of gun, it sends feedback to Arduino and the

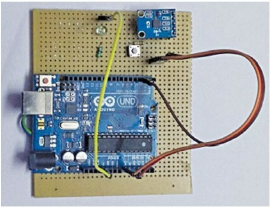
Fig. 4: Author's prototype

byte is changed back to 0. Frames of birds are taken from a .GIF file, and these are loaded one by one on the screen to show motion of the birds.

In the end, total score is shown on the screen along with the time taken and the shooting accuracy. It is possible to kill two or more coincident birds at one shot, and extra points are awarded for this.

Arduino code. Software (duck_hunt.ino) is written in Arduino programming language. Arduino Uno Board1 is programmed using Arduino IDE software. Connect Arduino board to the PC and select the correct COM port in Arduino IDE. Compile the program (sketch). Select the correct board from Tools→Board menu in Arduino IDE and upload the sketch.

Steps for testing

Step 1. Download Python 3.4.4 from its website. Select 64- or 32-bit software based on your Windows PC configuration. Install it.

Step 2. Go to command prompt Start→Run. Type cmd, and change your folder directory (where Python was installed). In my case it was in C drive, so I typed 'cd C:\Python34'.

Step 3. To avoid errors, update pip module type python -m pip install pip –upgrade. This will download the latest version of pip.

Step 4. Download Python packages from *www.lfd.uci.edu/~gohlke/pythonlibs/*. Search for pyserial and pygame (these two packages are required) from the list and download cp34 version or the latest one in pyserial. Also, download amd64 if your Windows is 64-bit or else download the 32-bit version.

Step 5. Save .whl files in a folder and install using pip. Type 'C:\Python34\Scripts\pip'. Install [[PATH]]\[[FILENAME]].whl. And, import the package in Python shell. It should work properly.

Step 6. After installing Python and its libraries, test Python code. It is simple—code given with a set of images has to be kept in the same folder. Open the code; there is a variable called PORT. Change that COM port number as per your PC, after you have attached, Arduino board.

Step 7. Attach the USB cable and after uploading the code in Arduino, run Python code by pressing F5. This will open a GUI with four birds flying around on the screen and a scope view of a gun as shown in Fig. 1.

Step 8. The gun's point can be changed by tilting the hardware, and bullets can be fired using S1.

Step 9. When you shoot while moving, the aim would not be accurate. After killing four birds, if time limit (150 seconds) is over, status of your game will be displayed on the screen. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*


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